



Application of polyvinyl alcohol/polypropylene/zinc oxide nanocomposites as sensor: modeling approach

Rania Badry¹ · Ahmed Fahmy¹ · Asmaa Ibrahim² · Hanan Elhaes¹ · Medhat Ibrahim³

Received: 3 April 2020 / Accepted: 20 November 2020 / Published online: 4 January 2021
© Springer Science+Business Media, LLC, part of Springer Nature 2021

Abstract

The blending process for polymer derivatives makes it suitable for providing superior properties to more economic materials. In addition, the importance of the blending process increased with the addition of nano scale metal oxide (MO), due to their versatile applications in gas sensors, biosensor and solar cells. As a result of blending, many important properties of the materials are changed, including electronic properties, thermal properties such as enthalpy, entropy, heat capacity and final heat of formation and surface properties. This leads to continuous experimental and theoretical monitoring of the materials used for blending. A theoretical study on the electronic and thermal properties of polyvinyl alcohol (PVA) blended with polypropylene (PP) was therefore conducted using the functional density theory DFT: B3LYP/ LANL2DZ. Results indicate PVA/PP is becoming more reactive. The addition of zinc oxide (ZnO) greatly enhances the electronic characteristics of the structures studied and confirms their applicability in the production of sensing devices. It was found that, as an adsorbed state, H₂S interacted with 4PVA/4PP/4OZn blend, the TDM increased to 10.0896 Debye and HOMO/LUMO band gap energy decreased to 0.6065 eV indicating the suitability of the presented blend for H₂S gas sensing.

Keywords Metal oxide · Electronic and thermal properties · PVA · PP · B3LYP/ LANL2DZ

1 Introduction

Polyvinyl Alcohol abbreviated as PVC and PVA is a polymer that is simple and easy to handle. (D'Aquino et al. 2012; Al-Malack 2001; Rajendran and Uma 2000). According to its abundance, it is dedicated for many applications in many areas (Roohani et al. 2008; Sun

✉ Medhat Ibrahim
medahmed6@yahoo.com

¹ Physics Department, Faculty of Women for Arts, Science and Education, Ain Shams University, Cairo 11757, Egypt

² Basic Science Department, October High Institute For Engineering & Technology, 6th October City, Giza, Egypt

³ Molecular Spectroscopy and Modeling Unit, Spectroscopy Department, National Research Centre, 33 El-Bohouth St, Dokki, Giza 12622, Egypt

et al. 2016; Brydson 1995). As previously stated, it is a hot topic of research to improve its functionality, depending on its low price and easy handling. (Chopra 2017; Mallakpour and Shafiee 2017). This leads to continuous interest in research on both experimental and theoretical bases (Kim 2001). Nano scale material interaction with PVC leads to nanocomposites of great concern as sensors, especially for gas sensors (Hoherčáková and Opekar 2004; Hu 2012). PP is non-polar and semicrystalline, and is an attractive option to use as low-cost commercial polyolefins. It is also attributed to it as lightweight, easy to process and has a wide range of properties and applications (Kumar and Gupta, 2018; El Achaby et al. 2012; Świergiel and Jadżyn 2020). Investigation into the physical properties of polymers (natural and/or synthetic) and blends of polymers could be studied effectively with molecular modeling (Ibrahim and Osman 2009; Badry et al. 2018; Ibrahim et al. 2019, 2010). In this sense, molecular modelling could investigate the total dipole moment (TDM) and the HOMO / LUMO band gap energy that all these physical properties are an indicator of a given structure's reactivity (Ibrahim et al. 2005). The most important factor is that with molecular electrostatic potential (MESP), surface-active sites of the polymers studied could be achieved as a result of interaction with a given gas. Mapping electrostatic potential is a good surface indicator for studying surface that acts as gas sensor. Polymers such as PVC and PVA could interact with MO, and the calculated physical parameters with molecular modeling indicated that this class of nano composites could be used as sensors (Fahmy et al. 2020; Alghunaim et al. 2017).

PVC in the environment could be a source of hazardous material such as chlorinated hydrocarbons which are not environmentally degradable and considered hazardous products (Czégény et al. 2012; Kulesza and German 2003). PVC could also produce chlorinated hydrocarbons during the co-pyrolysis process (Dai et al. 2018). Attempts to have similar properties to PVC achieved with polyvinyl alcohol PVA, which is less toxic and water soluble (Rudra et al. 2015). In addition, hydroxyl groups improve PVA's ability to further interact and increase the electronic properties. This paves the way for emerging PVA applications in many fields, especially in biomedical applications as well as in many other important applications. (Paralikar et al., 2008). In this sense PVA is an interesting choice for further blending with PP and MO. In general, it could be concluded that electronic structure methods could be important to understand the functionality of polymers (Elhaes et al. 2014; Omara et al. 2015). As far as certain gas interacts with a given surface, there may be changes in the physical properties of the surface which could in turn generate a signal that could be amplified. Such surface acts as sensor to this gas (Wang et al. 2010; Yuan and Shi 2013; Liu et al. 2012; Mao et al. 2014). The most important aspects are to design of an easy to handled surface, with good response and low cost. Molecular modeling is one of the promising tools to design such surfaces based on calculations of the surface's physical properties and to describe the interaction mechanism between the surface and the studied gas. In this sense, the present work is dedicated to study PVA/PP/ZnO as sensor for H₂S at B3LYP/ LANL2DZ level of theory.

2 Calculation details

All calculations were carried out on personal computer using GAUSSIAN 09 program packages at Spectroscopy Department, National Research Centre (Frisch et al. 2010). The studied structures of Polyvinyl alcohol (PVA), Polypropylene (PP), Zinc Oxide (ZnO), PVA/PP and PVA/PP/ZnO were optimized using B3LYP/LANL2DZ model (Becke 1993,

Fig. 1 Two schemes for the prepared model molecules to act as sensor namely PVA/PP/ZnO and PVA/PP/OZn

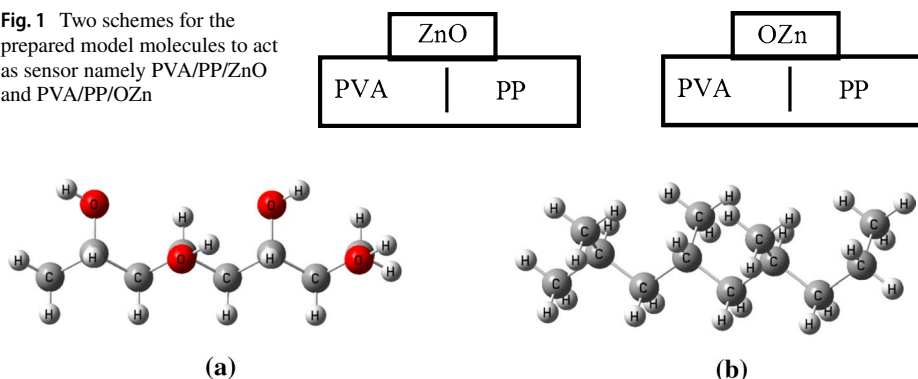


Fig. 2 The B3LYP/LANL2DZ optimized structure of **a** emeraldine base PVA and **b** emeraldine base PP

1996; Lee et al. 1988). The frontier molecular orbitals energy (HOMO/LUMO), total dipole moment (TDM) and molecular electrostatic potential (MESP) were also calculated with the same model. The enthalpy, entropy, heat capacity and final heat of formation were calculated with semi-empirical PM6 quantum mechanical calculations using SCIGRESS 3.0 program (Stewart 2008).

3 Results and discussion

Construction of the model molecule for sensing applications. As indicated in Fig. 1, there are two configurations: the first one is to prepare PVA/PP/ZnO blend while the second is PVA/PP/OZn. After performing molecular modeling for different configuration, the best configuration is chosen based on the studied physical parameters and then the sensing gas molecule H_2S will be further interacted.

It is well known that, increasing TDM with decreasing HOMO/LUMO band gap energy is an indication for the reactivity of a given compound (Ibrahim and Elhaes 2005; Ibrahim 2009; Ibrahim and Mahmoud 2009). On the other hand, MESP reflects the active sites of a given structure which is also a measure of reactivity of a surface of the given structure. It describes the interaction in terms of the transfer of charge in and out to the surface (Politzer et al. 1985, 2009). For a structure to act as sensor it is important to describe this structure in terms of TDM, HOMO/LUMO and MESP. Furthermore, the good stability of the structure could be indicated in terms of the calculated thermochemical parameters.

3.1 Interaction of PVA with PP

The blending process of PVA and PP was supposed to be heterogeneous polymer blend. Which means that the monomer unit consists of two different materials described by the formula $-(PVA-PP)_n-$. The proposed models which represent PVA and PP in their emeraldine shape were presented in Fig. 2.

The interaction between PVA and PP was supposed to be via C–C which is presented in Fig. 3. Also, the same interaction mechanism was supposed in the case of increasing the number of units to four units.

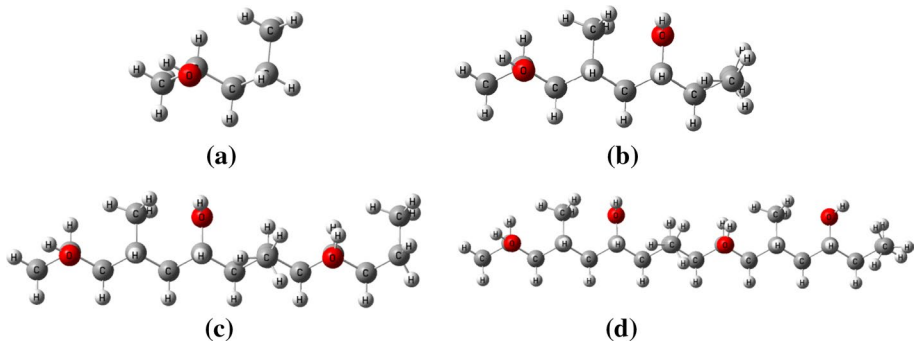


Fig. 3 The B3LYP/LANL2DZ optimized structure of **a** 1PVA/1PP; **b** 2PVA/2PP; **c** 3PVA/3PP and **d** 4PVA/4PP

3.2 Interaction of PVA/ PP blend with ZnO

ZnO is n-type semiconductor widely used as gas sensor as it is characterized by its low cost, high electron mobility and wide band gap, which dedicate it for detecting many gases (Şahin et al. 2015; Gong et al. 2006; Krishnakumar et al. 2009). Due to the favorable characteristics of 4PVA/4PP heterogeneous polymer blend, ZnO was added to this blend. The interaction between the blend and ZnO was done via the hydroxyl group (OH) of PVA through weak interaction. The effect of ZnO addition on the 4PVA/4PP heterogeneous blend was studied also by increasing the number of ZnO units from one unit to 4 units. The scheme presenting the interaction of different number of ZnO with 4PVA/4PP blend was presented in Fig. 4. The interaction between 4PVA/4PP was supposed to be Van der Waals interaction and a hydrogen bond was formed between the 4PVA/PP model molecule and ZnO.

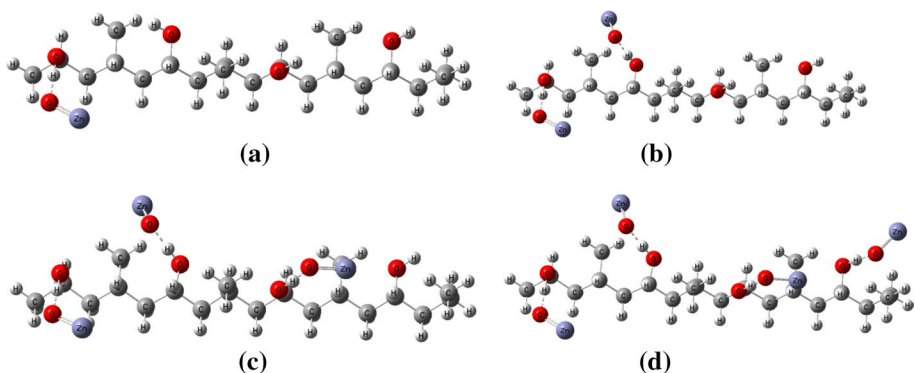


Fig. 4 The B3LYP/LANL2DZ optimized structure of 4PVA/4PP doped with **a** 1OZn; **b** 2OZn; **c** 3OZn and **d** 4OZn

Table 1 TDM and energy gap (HOMO/LUMO) calculated at B3LYP/LANL2DZ and enthalpy (En), entropy (Et), heat capacity (HC) and heat of formation (HF) calculated at PM6 semi-empirical level for XPVA/XPP interaction at 298 k where X = 1, 2, 3 and 4

Structure	TDM (Debye)	ΔE (eV)	En (Cal./mol)	Et (Cal/mol/K)	H.C (Cal/mol/K)	H.F (KCal/mol)
4PVA	3.1058	8.1463	10.807	134.339	58.719	-181.55
4 PP	0.1062	10.3668	11.332	136.315	63.666	-65.708
1 PVA-1PP	1.7822	9.3589	6.031	90.644	31.732	-73.830
2 PVA-2PP	3.4012	8.6498	11.018	133.750	61.397	-142.069
3 PVA-3PP	1.9245	8.4846	16.323	180.496	91.432	-210.083
4 PVA-4PP	1.0388	8.1183	21.480	224.026	121.271	-278.017

Table 2 TDM and energy gap (HOMO/LUMO) calculated at B3LYP/LANL2DZ and enthalpy (En), entropy (Et), heat capacity (HC) and heat of formation (HF) calculated at PM6 semi-empirical level for 4PVA/4PP/XOZn interaction at 298 k where X = 1, 2, 3 and 4

Structure	TDM (Debye)	ΔE (eV)	En. (Cal./mol)	Et. (Cal/mol/K)	H.C. (Cal/mol/K)	H.F. (KCal/mol)
4 PVA-4PP-1OZn	3.7235	2.1143	21.976	230.969	124.637	-300.696
4 PVA-4PP-2OZn	6.4132	1.9451	23.424	241.741	131.374	-335.557
4 PVA-4PP-3OZn	5.1487	1.9288	25.719	272.285	141.776	-360.085
4 PVA-4PP-4OZn	6.1908	1.9302	27.268	283.449	150.618	-388.653

3.3 Calculated physical parameters

Some parameters such as HOMO/LUMO band gap and TDM together with thermodynamic parameters such as enthalpy, entropy, heat capacity and heat of formation were calculated and tabulated in Tables 1 and 2 for the model molecules representing the different PVA/PP blends and for that representing PVA/PP/ZnO nanocomposite respectively.

The results indicated in Table 1 showed that 4PVA model molecule has a TDM of 3.1058 Debye while that 4PP model has a TDM of 0.1062 Debye. Upon blending the obtained values of TDM for 4PVA and 4PP become 1.7822, 3.4012, 1.9245 and 1.0388 eV for the different polymerization degrees 1PVA/1PP, 2PVA/2PP, 3PVA/3PP and 4PVA/4PP respectively. The changes occurred in the TDM values of PVA and PP due to blending indicated that the distribution of electrons changed within the proposed structure of PVA and PP.

Meanwhile, the calculated band gap energy of 4PVA and 4PP was changed from 8.1463 and 10.3668 eV respectively to 9.3589, 8.6498, 8.4846 and 8.1183 eV for 1PVA/1PP, 2PVA/2PP, 3PVA/3PP and 4PVA/4PP respectively. It is clear that, for PVA/PP polymer blends, the changes in the calculated values of both TDM and HOMO/LUMO band gap were relatively small.

Studying the thermal parameters helps in understanding the thermal behavior of various substances and their thermal stability. Where, enthalpy measures the total energy of the studied materials (i.e. PVA, PP and blended PVA/PP). Meanwhile, entropy measures the number of ways that the PVA and PP atoms were arranged. However, the amount of energy

needed to raise the temperature of PVA and PP by unity is called the heat capacity. Finally, other thermal parameters such as heat of formation was calculated to express the amount of energy released or absorbed when the studied molecule is formed.

Table 1 presents the calculated values of enthalpy, entropy, heat capacity and heat of formation for 4PVA, 4PP and XPVA/XPP blends where $X=1, 2, 3$ and 4 at PM6 semi-empirical level. Also, these thermal parameters were calculated for all structures at different temperatures ranging from 200 up to 500 K as presented in Fig. 5. The results indicated that by increasing the polymerization degree to 2, 3 and 4, the enthalpy of PVA and PP increased to 11.018, 16.323 and 21.480 Cal./mol for 2PVA/2PP, 3PVA/3PP and 4PVA/4PP respectively as tabulated in Table 1. The effect of blending on the enthalpy of PVA and PP was presented in Fig. 5a. Where, with increasing the temperature, the enthalpy increased. Meanwhile, the entropy of PVA and PP was changed from 134.339 and 136.315 Cal/mol/K respectively to 90.644, 133.750, 180.496 and 224.026 Cal/mol/K for 1PVA/1PP, 2PVA/2PP, 3PVA/3PP and 4PVA/4PP respectively. The influence of increasing the temperature from 298 to 500 K upon entropy was presented in Fig. 5b. Where, the figure indicated that the entropy of the blended polymers increased with increasing both temperature and polymerization degree. Additionally, heat capacity and the final heat of formation were also affected by increasing the degree of polymerization and with increasing temperature. Where, the PVA and PP heat capacity changed from 58.719 and

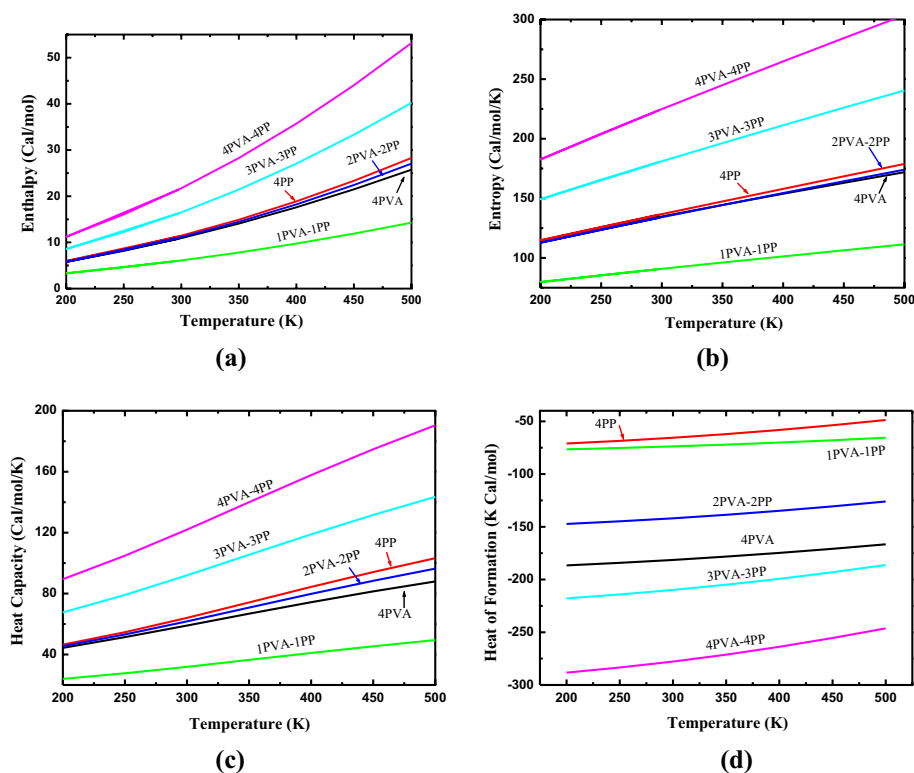


Fig. 5 The PM6 calculated **a** enthalpy; **b** entropy; **c** heat capacity and **d** heat of formation as a function of temperature for 4PVA, 4PP, 1PVA-1PP, 2PVA-2PP, 3PVA-3PP and 4PVA-4PP

63.666 Cal/mol/K to 31.732 Cal/mol/K for 1PVA/1PP, 61.397 Cal/mol/K for 2PVA/2PP, 91.432 Cal/mol/K for 3PVA/3PP and 121.271 Cal/mol/K for 4PVA/4PP. Meanwhile, the effect of changing the polymerization degree on the final heat of formation was changed from -181.55 kcal/mol for 4PVA and -65.708 kcal/mol for 4PP to -73.830 kcal/mol for 1PVA/1PP, -142.069 kcal/mol for 2PVA/2PP, -210.083 kcal/mol for 3PVA/3PP and -278.017 kcal/mol for 4PVA/4PP. However, the effect of temperature on both heat capacity and heat of formation was presented in Figs. 5c, d respectively. Where, both quantities were increased with increasing temperature. But, increasing the polymerization degree results in increasing heat capacity and decreasing heat of formation.

As, HOMO/LUMO band gap of 4PVA and 4PP was decreased slightly due to blending, then higher levels of calculations can be started from this level by increasing MO content. The effect of MO addition such as ZnO to the blended 4PVA/4PP was also studied to predict the changes occurred in the physical properties of the 4PVA/4PP model molecule which calculated at B3LYP/LANL2DZ level. Figure 4 presents the optimized structures of 4PVA/4PP/ $X\text{ZnO}$ where $X = 1, 2, 3$ and 4. Table 2 presents the calculated values of TDM and HOMO/LUMO band gap energy of 4PVA/4PP/1ZnO, 4PVA/4PP/2ZnO, 4PVA/4PP/3ZnO and 4PVA/4PP/4ZnO. Where, TDM of 4PVA/4PP increased from 1.0388 Debye to 3.7235, 6.4132, 5.1487 and 6.1908 Debye for 4PVA/4PP/ 1ZnO, 4PVA/4PP/ 2ZnO, 4PVA/4PP/ 3ZnO and 4PVA/4PP/ 4ZnO respectively. The higher value of TDM of 4PVA/4PP reflects its reactivity and hence the ability to interact with other materials. Meanwhile, HOMO/LUMO band gap energy of 4PVA/4PP decreased from 8.1183 eV to 2.1143, 1.9451, 1.9288 and 1.9302 eV for 4PVA/4PP/ 1ZnO, 4PVA/4PP/ 2ZnO, 4PVA/4PP/ 3ZnO and 4PVA/4PP/ 4ZnO respectively. This sharp decrease in the 4PVA/4PP electronic energy gap confirmed the results of the calculated TDM that it can interact with the surrounding media.

Additionally, thermal parameters are also studied for this group of molecules to discuss their thermal stability. Figure 6 shows the changes occurred in enthalpy, entropy, heat capacity and heat of formation as a result of increasing temperature from 298 to 5 K 500 K. Where, Fig. 6a presents that the enthalpy of 4PVA and 4PP increased with increasing both temperature and OZn concentration. This increase in enthalpy values was presented in Table 2. Where, enthalpy of 4PVA and 4PP at 298 K increased to 21.976, 23.424, 25.719 and 27.268 Cal/mol for 4PVA/4PP/1OZn, 4PVA/4PP/2OZn, 4PVA/4PP/3OZn and 4PVA/4PP/4OZn respectively. Also, entropy of 4PVA and 4PP was increased linearly with temperature as presented in Fig. 6b. The entropy of 4PVA and 4PP, at room temperature, was changed to 230.969 Cal/mol/K for 4PVA/4PP/1OZn, 241.741 Cal/mol/K for 4PVA/4PP/2OZn, 272.285 Cal/mol/K for 4 PVA/4PP/3OZn and 283.449 Cal/mol/K for 4 PVA/4PP/4OZn. Also, the heat capacity follows the same behavior of enthalpy and entropy where it increased with increasing temperature and OZn content. Where, it takes values of 124.637, 131.374, 141.776 and 150.618 Cal/mol/K respectively as seen in Fig. 6 c. However, the final heat of formation follows an inverse behavior compared to that of heat capacity as seen in Fig. 5d. Where, heat of formation decreased due to increasing OZn concentration as it decreased to -300.696 , -335.557 , -360.085 and -388.653 K Cal/mol as presented in Table 2.

Based on the results obtained, it was concluded that the model molecules based on PVA/PP/MO are more reactive than that of PVA/PP. Therefore, another important physical descriptor was also studied for PVA/PP/OZn model molecules which is the MESP. The importance of electrostatic potential lies in its ability to describe the surface of the studied structures i.e. the active sites present. Where, MESP defines the active sites in the molecule by following a color scheme starts from blue and ends at red ongoing from positive regions

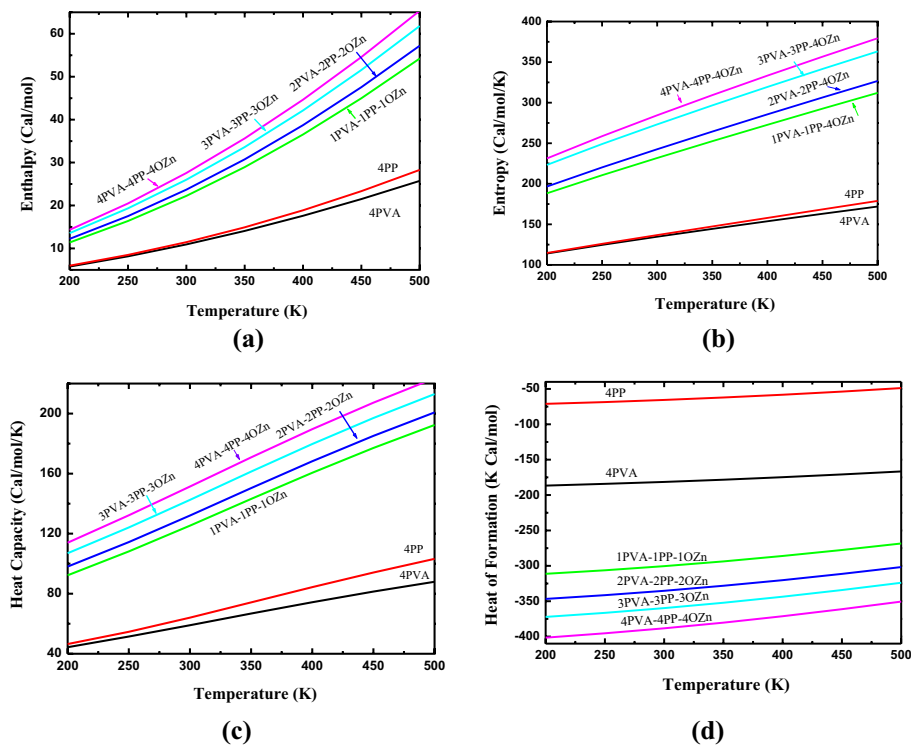


Fig. 6 The PM6 calculated **a** enthalpy; **b** entropy; **c** heat capacity and **d** heat of formation as a function of temperature for 4PVA-4PP-1ZnO, 4PVA-4PP-2ZnO, 4PVA-4PP-3ZnO and 4PVA-4PP-4ZnO

(regions that haven't electrons) to negative regions (i.e. regions with very high electro-negativity). The red regions refer to the active sites within the studied molecules. Figure 7 shows the MESP calculated as total density for 4PVA, 4PP and blended 4PVA/4PP with different concentrations of ZnO. As shown in Fig. 7 the red color was present only in the regions surrounding the OH group in the 4PVA structure. While in the 4PP structure there were no charged regions since only the blue color present within the molecule. However, with the addition of 1, 2, 3 and 4 units of OZn the intensity of the red color increased throughout the structure of 4PVA/4PP and its noticed from the figures that the red color localized around the OZn and OH group. These results confirmed that the reactivity of 4PVA and 4PP were increased strongly due to blending and metal oxide addition.

3.4 Sensitivity of 4PVA/4PP/4OZn to H₂S gas

As MOs have a number of features such as low cost, simple measurement and easy production. MOs were extensively used as gas sensors. So that, the addition of nano MOs to a sensing material improves the gas sensing properties of such material. Accordingly, we can improve the sensitivity of 4PVA/4PP/4OZn model molecule via the interaction with hydrogen sulfide (H₂S) gas. H₂S gas was a flammable, colorless, corrosive and poisonous gas generally found during the production and drilling processes of natural gas and crude oil. Also, it is found in wastewater treatment and sewers.

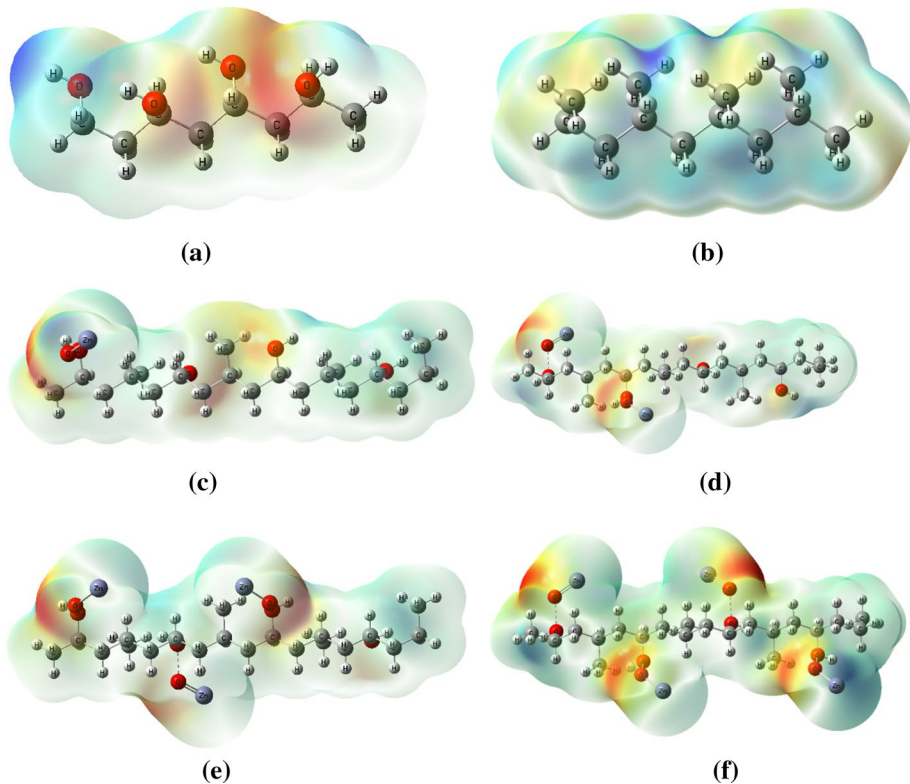


Fig. 7 The B3LYP/LANL2DZ calculated MESP as total density for **a** 4PVA; **b** 4PP; **c** 4PVA/4PP/1ZnO; **d** 4PVA/4PP/2ZnO; **e** 4PVA/4PP/3ZnO and **f** 4PVA/4PP/4ZnO

It is stated earlier that, ZnO has been proven to serve as a scavenger for H_2S gas in a variety of fluids and gases, particularly hydrocarbon gases containing H_2S or other sulfur-containing compounds and industrial flue gases (Hartmann 2007; Johnstone and Singh 1940).

In spite of this the present computational work suppose that the ZnO is impeded in nano scale into the studied composite, so that it may be show abnormal behavior according to the occurrence in nanoscale.

Here, the interaction between 4PVA/4PP/4OZn and H_2S was supposed to be only weak interaction and the H_2S gas was interacted with the polymer blend model of 4PVA/4PP/4OZn through the oxygen atom of PVA. The calculations were performed at B3LYP/LANL2DZ level of DFT. Figure 8 presents the studied model of H_2S gas and 4PVA/4PP/4OZn/ H_2S .

The results were indicated in terms of two of the most useful physical parameters used in describing the sensitivity of unknown materials. Such parameters were the TDM and the HOMO/LUMO band gap energy.

Due to the interaction of H_2S gas with 4PVA/4PP/4OZn throughout the oxygen atom of PVA, H_2S atoms effect on the TDM and HOMO/LUMO band gap energy. One can see, from Table 3, that TDM is increased from 6.1908 to 10.0896 Debye corresponding to the interaction of H_2S gas with the studied model of PVA/PP/OZn. While, the band gap energy decreased

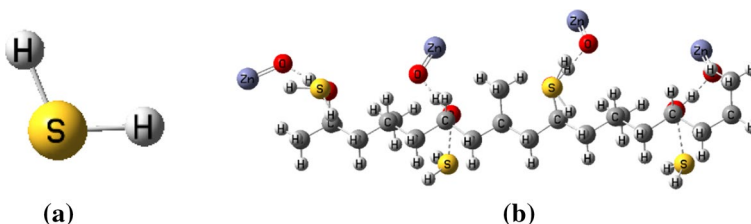


Fig. 8 The B3LYP/LANL2DZ optimized structure of **a** H_2S gas and **b** 4 PVA-4PP-4OZn-4 H_2S

Table 3 TDM and energy gap (HOMO/LUMO) calculated at B3LYP/LANL2DZ of H_2S and the blended 4 PVA/4PP/4OZn interacted with 4 H_2S

Structure	TDM (Debye)	ΔE (eV)
H_2S	1.9392	8.1923
4 PVA-4PP-4OZn-4 H_2S	10.0896	0.6065

from 1.9302 to 0.6065 eV. Although, H_2S gas has TDM of 1.9392 Debye and HOMO/LUMO band gap energy of 8.1923 eV as presented in Table 3.

The changes occurred in the studied physical properties of 4PVA/4PP/4OZn model molecule as its surface was exposed to H_2S gas was considered an indication for the sensitivity of the 4PVA/4PP/4OZn surface for H_2S gas. It is concluded, based on these considerations, that the blended 4PVA/4PP/4OZn structure could be used as a gas sensor for H_2S .

4 Conclusion

Some electronic, thermal and surface properties were studied for different number of heterogeneous PVA/PP polymer blend and PVA/PP/XOZn nanocomposites. Calculated TDM and band gap energy reflect the fact that the supposed structures of PVA/PP blends were more reactive as compared with PVA and PP themselves. As TDM increased and HOMO/LUMO band gap energy decreased due to increasing the degree of polymerization and due to increasing the MO concentration.

Thermodynamic data agree very well with electrostatic potential results and revealed that the proposed structures of 4PVA and that of 4PP become more reactive due to the interaction with different concentrations of ZnO. Where enthalpy, entropy together with heat capacity were increased with increasing both temperature and ZnO units. Meanwhile, the final heat of formation decreased with increasing ZnO content. It could be concluded that, the interaction of H_2S gas increased the sensitivity of 4PVA/4PP/4OZn nanocomposites and that the response performance was improved. Where, TDM reached to 10.0896 Debye and the band gap energy decreased to 0.6065 eV. So that, it can be concluded that 4PVA/4PP/4OZn was a perfect sensor to H_2S gas.

References

- Alghunaim, N.S., Omar, A., Elhaes, H., Ibrahim, M.: Effect of ZnO and TiO₂ on the reactivity of some polymers. *J. Comput. Theor. Nanosci.* **14**, 2838–2843 (2017)
- Al-Malack, H.M.: Migration of lead from unplasticized polyvinyl chloride pipes. *J. Hazard Mater.* **82**, 263–274 (2001)
- Badry, R., Shaban, H., Elhaes, H., Refaat, R., Ibrahim, M.: Molecular modeling analyses of polyaniline substituted with alkali and alkaline earth elements. *Biointerface Res. Appl. Chem.* **8**, 3719–3724 (2018)
- Becke, A.D.: Density- functional thermo-chemistry III. The role of exact exchange. *J. Chem. Phys.* **98**, 5648–5652 (1993)
- Becke, A.D.: Density-functional thermochemistry IV. A new dynamical correlation functional and implications for exact-exchange mixing. *J. Chem. Phys.* **104**, 1040–1046 (1996)
- Brydson, J.A.: *Plastics Materials*, 6th edn. Butterworth-Heinemann, USA (1995)
- Chopra, S.: Electronic properties and optical absorption of graphene-polyvinylidene fluoride nanocomposites: A theoretical study. *Mater. Chem. Phys.* **186**, 159–166 (2017)
- Czégény, Z., Jakab, E., Blazsó, M., Bhaskar, T., Sakata, Y.: Thermal decomposition of polymer mixtures of PVC, PET and ABS containing brominated flame retardant: formation of chlorinated and brominated organic compounds. *J. Anal. Appl. Pyrol.* **96**, 69–77 (2012)
- Dai, H., Huang, Y., Huang, H.: Eco-friendly polyvinyl alcohol/carboxymethyl cellulose hydrogels reinforced with graphene oxide and bentonite for enhanced adsorption of methylene blue. *Carbohydr. Polym.* **185**, 1–11 (2018)
- D'Aquino, C.A., Balmant, W., Ribeiro, R.L.L., Munaro, M., Vargas, J.V.C., Amico, S.C.: A simplified mathematical model to predict PVC photodegradation in photobioreactors. *Polym. test.* **31**, 638–644 (2012)
- El Achaby, M., Arrakhiz, F.E., Vaudreuil, S., el KacemQuaiss, A., Bousmina, M., Fassi-Fehri, O.: Mechanical, thermal, and rheological properties of graphene-based polypropylene nanocomposites prepared by melt mixing. *Polym. Compos.* **33**, 733–744 (2012)
- Elhaes, H., Khafagy, M., Ibrahim, M., Abdel-Gawad, A.A.: Spectroscopic analyses of PVDX (X = F, Cl and Br). *J. Comput. Theor. Nanosci.* **11**, 2115–2119 (2014)
- Fahmy, A., Khafagy, R.M., Elhaes, H., Ibrahim, M.: Molecular Properties of Polyvinyl Alcohol/Sodium Alginate Composite. *Biointerface Res. Appl. Chem.* **10**, 4734–4739 (2020)
- Frisch, M.J., Trucks, G.W., Schlegel, H.B., Scuseri, G. E, Robb, M.A., Cheeseman, J.R., Scalmani, G., Barone, V., Mennucci, Petersson, B. G. A., Nakatsuji, H., Caricato, M., Li, X., Hratchian, P., Izmaylov, A.F., Bloino, J., Zheng, G., Sonnenberg, J.L., Hada, M., Ehara, M., Toyota, K., Fukuda, R., Hasegawa, J., Ishida, M., Nakajima, T., Honda, Y., Kitao, O., Nakai, H., Vreven, T., Montgomery, J.A., Peralta, J.E., Ogliaro, F., Bearpark, M., Heyd, J.J., Brothers, E., Kudin, K.N., Staroverov, V.N., Keith, T., Kobayashi, R., Normand, J., Raghavachari, K., Rendell, A., Burant, J.C., Iyengar, S.S., Tomasi, J., Cossi, M., Rega, N., Millam, J.M., Klene, M., Knox, J.E., Cross, J.B., Bakken, V., Adamo, C., Jaramillo, J., Gomperts, R., Stratmann, R.E., Yazyev, O., Austin, A.J., Cammi, R., Pomelli, C., Ochterski, J.W., Martin, R.L., Morokuma, K., Zakrzewski, V.G., Voth, G.A., Salvador, P., Dannenberg, J.J., Dapprich, S., Daniels, A.D., Farkas, O., Foresman, J.B., Ortiz, J.V., Cioslowski, J., Fox, D.J., Gaussian, Inc., Gaussian 09, Revision C.01. Wallingford CT (2010)
- Gong, H., Hu, J.Q., Wang, J.H., Ong, C.H., Zhu, F.R.: Nano-crystalline Cu-doped ZnO thin film gas sensor for CO. *Sens. Actuat. B- Chem.* **115**, 247–251 (2006)
- Hoherčáková, Z., Opekar, F.: Au/PVC composite a new material for solid-state gas sensors: Detection of nitrogen dioxide in the air. *Sens. Actuat. B.* **97**, 379–386 (2004)
- Hartmann, V.L.: Gas-solid reaction modeling as applied to the fine desulfurization of gaseous feedstocks. *Chem. Eng. J.* **134**, 190–194 (2007)
- Hu, J., Zhu, Y., Huang, H., Lu, J.: Recent advances in shape-memory polymers: Structure, mechanism, functionality, modeling and applications. *Prog. Polym. Sci.* **37**, 1720–1763 (2012)
- Ibrahim, A., Elhaes, H., Ibrahim, M., Yahia, I.S., Zahran, H.Y.: Molecular Modeling Analyses for Polyvinylidene X (X=F, Cl, Br and I). *Biointerface Res. Appl. Chem.* **9**, 3890–3893 (2019)
- Ibrahim, M., Elhaes, H.: Computational spectroscopic study of copper, cadmium, lead and zinc interactions in the environment. *Int. J. Environ. Pollut.* **23**, 417–424 (2005)
- Ibrahim, M., Mahmoud, A.A., Osman, O., Refaat, A., El-Sayed, E.M.: Molecular spectroscopic analyses of nano chitosan blend as biosensor. *Spectrochim. Acta A.* **77**, 802–806 (2010)
- Ibrahim, M., Mahmoud, A.-A.: Computational notes on the reactivity of some functional groups. *J. Comput. Theor. Nanosci.* **6**, 1523–1526 (2009)
- Ibrahim, M., Osman, O.: Spectroscopic analyses of cellulose: Fourier Transform infrared and molecular modelling study. *J. Comput. Theor. Nanosci.* **6**, 1054–1058 (2009)

- Ibrahim, M.: Molecular modelling and FTIR study for K, Na, Ca and Mg coordination with organic acid. *J. Comput. Theor. Nanosci.* **6**, 682–685 (2009)
- Johnstone, H.F., Singh, A.D.: Recovery of sulfur dioxide from waste gases. *Ind. Eng. Chem.* **32**, 1037–1049 (1940)
- Kim, S.: Pyrolysis kinetics of waste PVC pipe. *J. Waste Manag.* **21**, 609–616 (2001)
- Krishnakumar, T., Jayaprakash, R., Pinna, N., Donato, N., Bonavita, A., Micali, G., Neri, G.: CO gas sensing of ZnO nanostructures synthesized by an assisted microwave wet chemical route. *Sensor Actuat. B-Chem.* **143**, 198–204 (2009)
- Kulesza, K., German, K.: Chlorinated pyrolysis products of co-pyrolysis of poly(vinyl chloride) and poly(ethylene terephthalate). *J. Anal. Appl. Pyrol.* **67**, 123–134 (2003)
- Kumar, A., Gupta, R. K.: Fundamentals of polymer engineering. CRC Press (2018)
- Lee, C., Yang, W., Parr, R.G.: Development of the Colle-Salvetti correlation-energy formula into a functional of the electron density. *Phys. Rev. B.* **37**, 785–789 (1988)
- Liu, X., Cheng, S., Liu, H., Hu, S., Zhang, D., Ning, H.: A survey on gas sensing technology. *J. Sens.* **12**, 9635–9665 (2012)
- Mallakpour, S., Shafiee, E.: The synthesis of poly(vinyl chloride) nanocomposite films containing ZrO₂ nanoparticles modified with vitamin B1 with the aim of improving the mechanical, thermal and optical properties. *Des Monomers Polym.* **20**, 378–388 (2017)
- Mao, S., Lu, G., Chen, J.: Nanocarbon-based gas sensors: progress and challenges. *Mater. Chem. Phys. A.* **2**, 5573–5579 (2014)
- Omara, W., Amin, R., Elhaes, H., Ibrahim, M., Elfeky, S.A.: Preparation and characterization of novel poly-anilinenanosensor for sensitive detection of formaldehyde. *Recent Pat. Nanotech.* **9**, 195–203 (2015)
- Paralikar, S.A., Simonsen, J., Lombardi, J.: Poly(vinyl alcohol)/cellulose nanocrystal barrier membranes. *J. Membr. Sci.* **320**, 248–258 (2008)
- Politzer, P., Laurence, P.R., Jayasuriya, K.: Molecular electrostatic potentials: an effective tool for the elucidation of biochemical phenomena. *Environ. healthperspect.* **61**, 191–202 (1985)
- Politzer, P., Murray, J.S.: The electrostatic potential as a guide to molecular interactive behavior. In: PK Chattaraj, ed. *Chemical reactivity theory: A density functional view*. Boca Raton, FL: CRC Press; 243–254 (2009)
- Rajendran, S., Uma, T.: Effect of ZrO₂ on conductivity of PVC–LiBF₄–DBP polymer electrolytes. *Mater. Lett.* **44**, 208–214 (2000)
- Roohani, M., Habibi, Y., Belgacem, N.M., Ebrahim, G., Karimi, A.N., Dufresne, A.: Cellulose whiskers reinforced polyvinyl alcohol copolymers nanocomposites. *Eur. Polym. J.* **44**, 2489–2498 (2008)
- Rudra, R., Kumar, V., Kundu, P.P.: Acid catalysed cross-linking of poly vinyl alcohol (PVA) by glutaraldehyde: Effect of crosslink density on the characteristics of PVA membranes used in single chambered microbial fuel cells. *RSC Adv.* **5**, 83436–83447 (2015)
- Şahin, Z.S., Şenöz, H., Tezcan, H., Büyükgüngör, O.: Synthesis, spectral analysis, structural elucidation and quantum chemical studies of (E)-methyl-4-[(2-phenylhydrazono) methyl] benzoate. *Spectrochim. Acta A.* **143**, 91–100 (2015)
- Stewart, J.J.P.: MO-G Version 1.1A. Fujitsu Limited, Tokyo, Japan (2008)
- Sun, H., Cao, B., Tian, Q., Liu, S., Du, D., Xue, Z., Fu, H.: A DFT study on the absorption mechanism of vinyl chloride by ionic liquids. *J. Mol. Liq.* **215**, 496–502 (2016)
- Świergiel, J., Jadzyn, J.: A virtual transition from polypropylene glycol (PPG) to polypropylene oxide (PPO). *J. Mole. Liq.* **304**, 112748 (2020)
- Wang, C., Yin, L., Zhang, L., Xiang, D., Gao, R.: Metal oxide gas sensors: sensitivity and influencing factors. *J. Sens.* **10**, 2088–2106 (2010)
- Yuan, W., Shi, G.: Graphene-based gas sensors. *J. Mater. Chem. A.* **1**, 10078–10091 (2013)